By Qamar Raza

# LINUX BASIC COMMANDS

## ASSIGNMENT 1

### Section A: Navigation

1. **Create a directory named `linux_practice`.**

   ***mkdir linux_practice***

2. **Go inside that directory.**
   ***cd linux_practice***

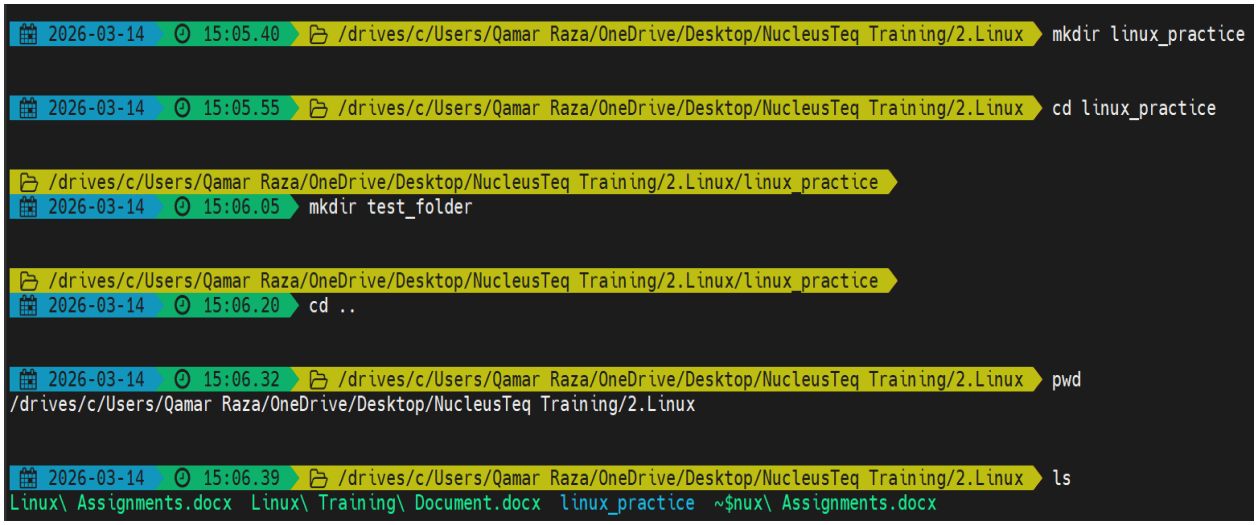
3. **Create another directory called `test_folder`.**
   ***mkdir test_folder***

4. **Move back to the parent directory.**
   ***cd ..***

5. **Display your current working directory.**
   ***pwd***

6. **List all files and directories.**
   ***ls***

7. **Find help for the `ls` command.**
   ***ls --help***



```
2026-03-14  15:06.46  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  ls --help
Toybox 0.8.10 multicall binary (see toybox --help)

usage: ls [-1ACFHLNRSUXZabcdfghilmnopqrstuwx] [--color[=auto]] [FILE...]

List files

what to show:
-A  all files except . and ..      -a  all files including .hidden
-b  escape nongraphic chars        -d  directory, not contents
-F  append /dir *exe @sym |FIFO    -f  files (no sort/filter/format)
-H  follow command line symlinks   -i  inode number
-L  follow symlinks                -N  no escaping, even on tty
-p  put '/' after dir names        -q  unprintable chars as '?'
-R  recursively list in subdirs    -s  storage used (in --block-size)
-Z  security context

output formats:
-1  list one file per line         -C  columns (sorted vertically)
-g  like -l but no owner           -h  human readable sizes
-l  long (show full details)       -ll long with nanoseconds (--full-time)
-m  comma separated                -n  long with numeric uid/gid
-o  long without group column      -r  reverse order
-w  set column width               -x  columns (horizontal sort)

sort by:  (also --sort=longname,longname... ends with alphabetical)
-c  ctime      -r  reverse    -S  size     -t  time    -u  atime    -U  none
-X  extension  -!  dirfirst   -~  nocase

--block-size N  block size (default 1024)
--color  =always (default)  =auto (when stdout is tty) =never
    exe=green  suid=red  suidfile=redback  stickydir=greenback
    device=yellow  symlink=turquoise/red  dir=blue  socket=purple

Long output uses -cu for display, use -ltc/-ltu to also sort by ctime/atime.

Important:
/bin/ls command is provided by "toybox" (http://landley.net/toybox)
▸ You can download the full genuine version by using the following command:
apt install coreutils grep tar xz cpio gawk nc sed findutils patch


2026-03-14  15:06.56  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux
```

## Section B: File & Directory Operations

1. **Create a file named `student.txt`.**

   ***touch student.txt***

2. **Create a directory called `data`.**
   ***mkdir data***

3. **Copy `student.txt` inside `data`.**
   ***cp student.txt data***

4. **Rename `student.txt` to `info.txt`.**
   ***mv student.txt info.txt***

5. **Delete the file `info.txt`.**
   ***rm info.txt***

6. **Remove the `data` directory using:**

   i) `rmdir`

   ***rmdir data***

   ii) `rm -rf`

   ***rm -rf data***

```
2026-03-14 15:06.56 /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux touch student.txt

2026-03-14 15:27.28 /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux mkdir data

2026-03-14 15:27.35 /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux cp student.txt data

2026-03-14 15:27.49 /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux mv student.txt info.txt

2026-03-14 15:28.06 /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux rm info.txt

2026-03-14 15:28.14 /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux rmdir data
rmdir: data: Directory not empty

2026-03-14 15:28.23 /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux rm -rf data

2026-03-14 15:28.33 /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux
```

# Assignment 2: File Editing & Text Processing

1. **Create a file called `bio.txt`.**

   *touch bio.txt*

2. **Add the following details using `vi`:**
   - **Your Name**
   - **Course Name**
   - **City**

   *vi bio.txt*

   (press "i" to switch to insert mode to write some content)

```
/NucleusTeq Training/2.Linux  touch bio.txt

/NucleusTeq Training/2.Linux  vi bio.txt
```

```
2. /drives/c/Users/Qar
Qamar Raza
Linux Operating System
Indore
~
```

3. **Save the file properly.**

   Press "Escape" button followed by a colon (:) followed by "wq!" and press "Enter"

```
:wq!
```

4. **Search for your name inside the file.**

   *grep "Qamar Raza" bio.txt*

```
p/NucleusTeq Training/2.Linux  grep "Qamar Raza" bio.txt
Qamar Raza
```

5. **Delete all content from the file using shortcut commands.**

   Open the file in vi editor using

   ***vi file_name.txt***

```
/NucleusTeq Training/2.Linux  vi bio.txt
```

   Now follow the following steps:

   - press gg

```
Qamar Raza
Linux Operating System
Indore
```

   - press VG

```
Qamar Raza
Linux Operating System
Indore
```

   - press dd

```
2. /drives/c/Use
~
```

6. **Use `grep` to:**

   - **Search your name in `bio.txt`**

     ***grep "Qamar Raza" bio.txt***

```
p/NucleusTeq Training/2.Linux  grep "Qamar Raza" bio.txt
Qamar Raza
```

   - **Show lines that DO NOT contain your name.**

     ***grep -v "Qamar Raza" bio.txt***

```
/NucleusTeq Training/2.Linux  grep -v "Qamar Raza" bio.txt
Linux Operating System
Indore
```

# Assignment 3: Linux Permissions

1. **Create a file `script.sh`.**

2. **Check its permission using `ls -l`.**

3. **Change permission to:**

   - **755**

   - **764**

```
2026-03-14  16:03.14  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  touch script.sh

2026-03-14  16:13.39  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  ls -l
total 642
-rwx------ 1 Qamar Raza  None  223606 2026-03-13 11:26 Linux\ Assignments.docx
-rwx------ 1 Qamar Raza  None 1087722 2026-03-13 11:29 Linux\ Training\ Document.docx
-rw----r-- 1 Qamar Raza  None      41 2026-03-14 15:55 bio.txt
drwx---r-x 1 Qamar Raza  None       0 2026-03-14 15:06 linux_practice
-rw----r-- 1 Qamar Raza  None       0 2026-03-14 16:13 script.sh
-rwx------ 1 Qamar Raza  None     162 2026-03-14 14:39 ~$nux\ Assignments.docx

2026-03-14  16:13.44  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  chmod 755 script.sh

2026-03-14  16:17.40  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  ls -l
total 642
-rwx------ 1 Qamar Raza  None  223606 2026-03-13 11:26 Linux\ Assignments.docx
-rwx------ 1 Qamar Raza  None 1087722 2026-03-13 11:29 Linux\ Training\ Document.docx
-rw----r-- 1 Qamar Raza  None      41 2026-03-14 15:55 bio.txt
drwx---r-x 1 Qamar Raza  None       0 2026-03-14 15:06 linux_practice
-rwxr-xr-x 1 Qamar Raza  None       0 2026-03-14 16:13 script.sh
-rwx------ 1 Qamar Raza  None     162 2026-03-14 14:39 ~$nux\ Assignments.docx

2026-03-14  16:17.44  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  chmod 764 script.sh

2026-03-14  16:18.11  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  ls -l
total 642
-rwx------ 1 Qamar Raza  None  223606 2026-03-13 11:26 Linux\ Assignments.docx
-rwx------ 1 Qamar Raza  None 1087722 2026-03-13 11:29 Linux\ Training\ Document.docx
-rw----r-- 1 Qamar Raza  None      41 2026-03-14 15:55 bio.txt
drwx---r-x 1 Qamar Raza  None       0 2026-03-14 15:06 linux_practice
-rwxrw-r-- 1 Qamar Raza  None       0 2026-03-14 16:13 script.sh
-rwx------ 1 Qamar Raza  None     162 2026-03-14 14:39 ~$nux\ Assignments.docx
```

4. **Explain what `764` means.**

   It is a 3 number code (XXX).

   The first number represents the access to the owner

   The second number represents the access to the group

   The third number represents the access to others

   Here:

   1st digit = 7: Owner can read(4) + write(2) + execute(1)

   2nd digit = 6: Group can read(4) + write(2)

   3rd digit = 4: Others can read(4)

5. **Give execute permission using symbolic method (`chmod +x`).**

   ***chmod +x script.sh***

```
2026-03-14  16:18.13  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  chmod +x script.sh

2026-03-14  16:21.28  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  ls -l
total 642
-rwx------ 1 Qamar Raza  None  223606 2026-03-13 11:26 Linux\ Assignments.docx
-rwx------ 1 Qamar Raza  None 1087722 2026-03-13 11:29 Linux\ Training\ Document.docx
-rw----r-- 1 Qamar Raza  None      41 2026-03-14 15:55 bio.txt
drwx---r-x 1 Qamar Raza  None       0 2026-03-14 15:06 linux_practice
-rwxrwxr-x 1 Qamar Raza  None       0 2026-03-14 16:13 script.sh
-rwx------ 1 Qamar Raza  None     162 2026-03-14 14:39 ~$nux\ Assignments.docx
```

6. **What is the meaning of r=4, w=2, x=1?**

   In Linux, permissions are based on a binary system. Each action has a unique numeric value so they can be added together without overlapping:

   - r (Read) = 4: Permission to open and view the file's contents.
   - w (Write) = 2: Permission to modify or delete the file.
   - x (Execute) = 1: Permission to run the file as a program or script.

   By using these specific numbers (powers of 2), any sum from 0 to 7 represents a unique combination of permissions. For example, a 5 can *only* be 4+1 (Read + Execute).

7. **What does permission 777 mean? Is it safe? Why?**

   Permission 777 means everyone (Owner, Group and Others) can perform all 3 operations (Read, Write and Execute) on a certain file.

   **Is it safe?** The answer is no, it is generally not safe.

   **Why?** Because without owner's permission, any one can edit the file content which is not ideal for almost every scenario.

# Assignment 4: Linux Networking

## Task 1: Check Network Configuration

1. **Check your system IP address.**

   ***Ifconfig***

```
2026-03-14  16:21.30  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  ifconfig

Software Loopback Interface 1
        Link encap: Local loopback
        inet addr:127.0.0.1 Mask: 255.0.0.0
        MTU: 1500 Speed:1073.74 Mbps
        Admin status:UP Oper status:OPERATIONAL
        RX packets:0 dropped:0 errors:0 unkown:0
        TX packets:0 dropped:0 errors:0 txqueuelen:0

Intel(R) Wi-Fi 6 AX101
        Link encap: IEEE 802.11 HWaddr: 28-0C-50-E8-8E-F1
        inet addr:192.168.1.5 Mask: 255.255.255.0
        MTU: 1500 Speed:411.65 Mbps
        Admin status:UP Oper status:OPERATIONAL
        RX packets:138560 dropped:0 errors:0 unkown:0
        TX packets:95553 dropped:0 errors:0 txqueuelen:0
```

2. **Write down about:**
   - **IP Address:**
     An **IP (Internet Protocol) Address** is a unique numerical label assigned to every device connected to a computer network. Think of it as a **digital home address** that allows devices to find and talk to each other.
     A standard IP address looks like this: 172.16.254.1.
     - It consists of **four numbers** separated by dots.
     - Each number can be anything from **0 to 255**.
     - The first part usually identifies the **network** you are on, and the last part identifies your **specific device**.

---

## Task 2: Test Network Reachability

3. *ping google.com*

```
2026-03-14  16:56.45  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  ping google.com

Pinging google.com [2404:6800:4002:828::200e] with 32 bytes of data:
Reply from 2404:6800:4002:828::200e: time=45ms
Reply from 2404:6800:4002:828::200e: time=48ms
Reply from 2404:6800:4002:828::200e: time=50ms
Reply from 2404:6800:4002:828::200e: time=46ms

Ping statistics for 2404:6800:4002:828::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 45ms, Maximum = 50ms, Average = 47ms
```

4. **Observe and write:**
   - **IP address resolved:**
     The resolved IP address is **2404:6800:4002:828::200e**
   - **Packet loss %:**
     The packet loss is **0%.** This means that connection is stable
     And all 4 packets sent were successfully received back

---

## Task 3: Check Port Connectivity

5. **Check if port 80 is open on `google.com` using telnet:**
   ***telnet google.com 80***

```
/NucleusTeq Training/2.Linux  telnet google.com 80
```

```
Connected to google.com.
```

6. **What does it mean if:Download `google.com` homepage.**
   - **Connection successful?**
     The request reached the server, and the server said "Hello" back. The handshake worked, and data is flowing

   - **Connection refused?**
     The target host is active, but it is explicitly rejecting the connection on that specific port.

   - **Connection timed out?**
     The request was sent, but no response was received within the allowed timeframe, often due to network issues or a silent firewall.

## Task 4: Check Port Connectivity

7. **Save it into a file.**
   *curl google.com > my_file.txt*

8. **Verify the file is created.**
   *ls -l my_file.txt*

```
2026-03-14  17:35.22  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  curl google.com > my_file.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   219  100   219    0     0    376      0 --:--:-- --:--:-- --:--:--   376


2026-03-14  17:36.04  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  ls -l my_file.txt
-rw----r-- 1 Qamar Raza  None 219 2026-03-14 17:36 my_file.txt


2026-03-14  17:36.10  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux
```

# Assignment 5: Process & System Monitoring

1. **List all running processes.**

***ps aux***

```
2026-03-14  17:36.10  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  ps aux
PID     PPID    USER        COMMAND
0       0       Unknown     [System Process]
4       0       Unknown     System
236     4       Unknown     Secure System
280     4       Unknown     Registry
796     4       Unknown     smss.exe
1108    944     Unknown     csrss.exe
1196    944     Unknown     wininit.exe
1340    1196    Unknown     services.exe
1364    1196    Unknown     LsaIso.exe
1372    1196    Unknown     lsass.exe
1516    1340    Unknown     svchost.exe
1548    1340    Unknown     WUDFHost.exe
1556    1196    Unknown     fontdrvhost.exe
1676    1340    Unknown     svchost.exe
1724    1340    Unknown     svchost.exe
1812    1340    Unknown     WUDFHost.exe
1876    1340    Unknown     svchost.exe
1896    1340    Unknown     svchost.exe
2040    1340    Unknown     svchost.exe
1188    1340    Unknown     WUDFHost.exe
1404    1340    Unknown     svchost.exe
2068    1340    Unknown     svchost.exe
2076    1340    Unknown     svchost.exe
2088    1340    Unknown     svchost.exe
```

2. **Open `top` command and monitor CPU usage.**

```
2026-03-14  17:43.03  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  top
Process Name        PID  PPID  CPU Mem(KB) IO/s Prio Threa Started Owner        Command
System                4     0   6%     124    0    8   316 26-03-13 Unknown      NULL
Secure System       236     4   0%   62944    0    8     0 26-03-13 Unknown      NULL
Registry            280     4   0%   57692    0    8     4 26-03-13 Unknown      NULL
smss                796     4   0%     748    0   11     2 26-03-13 Unknown      NULL
csrss              1108   944   0%    5032    0   13    12 26-03-13 Unknown      NULL
wininit            1196   944   0%    4988    0   13     2 26-03-13 Unknown      NULL
services           1340  1196   0%   11676    0    9    10 26-03-13 Unknown      NULL
LsaIso             1364  1196   0%    1820    0    8     1 26-03-13 Unknown      NULL
lsass              1372  1196   1%   26640    0    9    10 26-03-13 Unknown      NULL
svchost            1516  1340   0%   33144    0    8    18 26-03-13 Unknown      NULL
WUDFHost           1548  1340   0%   11520    0   13    17 26-03-13 Unknown      NULL
fontdrvhost        1556  1196   0%    1448    0    8     5 26-03-13 Unknown      NULL
svchost            1676  1340   1%   16860    0    8    13 26-03-13 Unknown      NULL
svchost            1724  1340   0%   10632    0    8     6 26-03-13 Unknown      NULL
WUDFHost           1812  1340   0%    6196    0    8    10 26-03-13 Unknown      NULL
svchost            1876  1340   0%    3016    0    8     3 26-03-13 Unknown      NULL
svchost            1896  1340   0%    9872    0   10     5 26-03-13 Unknown      NULL
svchost            2040  1340   0%    6708    0    8     7 26-03-13 Unknown      NULL
WUDFHost           1188  1340   0%    3556    0    8     5 26-03-13 Unknown      NULL
svchost            1404  1340   0%    8040    0    8    10 26-03-13 Unknown      NULL
svchost            2068  1340   0%    7556    0    8     5 26-03-13 Unknown      NULL
svchost            2076  1340   0%    8044    0    8     3 26-03-13 Unknown      NULL
svchost            2088  1340   0%   13124    0    8    10 26-03-13 Unknown      NULL
svchost            2192  1340   0%    6676    0    8     4 26-03-13 Unknown      NULL
IntelCpHDCPSvc     2236  1340   0%    2876    0    8     3 26-03-13 Unknown      NULL
svchost            2248  1340   0%    6244    0    8     9 26-03-13 Unknown      NULL
svchost            2308  1340   0%    3228    0    8    14 26-03-13 Unknown      NULL
svchost            2340  1340   0%   16792    0    8    12 26-03-13 Unknown      NULL
```

3. **Check disk usage.**

   ***df -h***

```
2026-03-14  17:48.38  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  df -h
Filesystem      Size Used Avail Use% Mounted on
C:              231G  90G  141G  40% /drives/c
D:              245G 2.6G  242G   2% /drives/d
```

4. **Check memory usage.**

   ***free -h***

```
2026-03-14  17:48.59  /drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux  free -h
                 total        used        free      shared     buffers
Mem:               16G        9.0G        6.7G           0           0
-/+ buffers/cache:            9.0G        6.7G
Swap:            2.3G        110M        2.2G
```

5. **Difference between ps and top**

   **ps** :- It stands for "Process Status". It is a static snapshot of what is running at the exact millisecond you press Enter. It gives you the information and immediately returns you to the command prompt.

   **top** :- It stands for "Table of Processes". It is a live, real-time dashboard. It stays open and refreshes (usually every 3 seconds) to show you how CPU and Memory usage change over time.

| Feature | ps | top |
|---|---|---|
| View | Static (Fixed) | Dynamic (Live) |
| Interactivity | None | High (You can sort, kill or filter inside it) |
| Resource Use | Very low | Higher (Because it constantly refreshes) |
| Output | Prints to the terminal and stops | Takes over the terminal window |

**What does df -h show?**

It stands for DISK FREE – HUMAN READABLE. It shows you how much space is left on your hard drives. (Check Assignment 5.3)

# Assignment 6: Shell Scripting (Basic)

## Task 1: Simple Script

Write a shell script that:

- Prints "Welcome to Linux Training"
- Prints current date
- Prints current user name

PROGRAM:

```
#!/bin/bash
echo "Welcome to Linux Training"
date
whoami
```

```
#!/bin/bash
echo "Welcome to Linux Training"
date
whoami
```

OUTPUT:

```
NucleusTeq Training/2.Linux  ./task1.sh
```

```
Welcome to Linux Training
Sat Mar 14 23:15:13 IST 2026
Qamar Raza
```

## Task 2: User Input Script

Create a script that:

- Takes user name as input
- Prints: "Hello <name>, welcome to Linux"

PROGRAM:

- ***read -p "Enter username: " username***
- ***echo "Hello $username, welcome to Linux"***

```
#!/bin/bash
read -p "Enter username: " username
echo "Hello $username, welcome to Linux"
```

OUTPUT:

```
/drives/c/Users/Qamar Raza/OneDrive/De
2026-03-14  23:23.42  ./task2.sh
Enter username: Qamar Raza
Hello Qamar Raza, welcome to Linux
```

---

# Conditional & Loop Script

## Task 3: Even or Odd Program

Write a script that:

- Takes a number as input
- Checks whether number is lesser than 10 or not

PROGRAM:

- *#!/bin/bash*
- *echo "Enter any number"*
- *read number*
- *if [ $number -ge 10 ]*
- *then echo "The number $number is greater than or equal to 10"*
- *else echo "The number $number is smaller than 10"*
- *fi*
- 
- *if [ $((number % 2)) -eq 0 ]*
- *then echo "The number $number is even!"*
- *else echo "Also the number $number is odd!"*
- *fi*

```
#!/bin/bash
echo "Enter any number"
read number
if [ $number -ge 10 ]
then echo "The number $number is greater than or equal to 10"
else echo "The number $number is smaller than 10"
fi

if [ $((number % 2)) -eq 0 ]
then echo "The number $number is even!"
else echo "Also the number $number is odd!"
fi
```

OUTPUT:

```
/drives/c/Users/Qamar Raza/OneDrive/Desktop
2026-03-14  23:35.48  ./task3.sh
Enter any number
9
The number 9 is smaller than 10
Also the number 9 is odd!
```

## Task 4: Loop Program

Write a script that:

- Prints numbers from 1 to 10
- Prints multiplication table of 5

PROGRAM:

- *echo "Numbers from 1 to 10:"*
- *for i in {1..10}*
- *do echo "$i"*
- *done*
- *echo "Multiplication table of 5:"*
- *for i in {1..10}*
- *do echo "5 X $i = $((5 * i))"*
- *done*

```
echo "Numbers from 1 to 10:"
for i in {1..10}
do echo "$i"
done

echo "Multiplication table of 5:"
for i in {1..10}
do echo "5 X $i = $((5 * i))"
done
```

OUTPUT:

```
2026-03-14  23:46.28  ./task4.sh
Numbers from 1 to 10:
1
2
3
4
5
6
7
8
9
10
Multiplication table of 5:
5 X 1 = 5
5 X 2 = 10
5 X 3 = 15
5 X 4 = 20
5 X 5 = 25
5 X 6 = 30
5 X 7 = 35
5 X 8 = 40
5 X 9 = 45
5 X 10 = 50
```

# Assignment 7: Mini Project

Create a script that:

1. Creates a directory `backup`
2. Takes current date and time
3. Copies all `.txt` files into backup folder
4. Prints "Backup Completed"
5. If backup folder already exists, print "Directory already exists"

**PROGRAM:**

```
#!/bin/bash

if [ -d backup ]

then

echo “Directory already exists!”

else

mkdir backup

date=$(date + “%Y-%m-%d_%H-%M-%S”

echo “Backup completed”
```

```
#!/bin/bash
if [ -d backup ]
then
echo "Directory already exists!"
else
mkdir backup
fi

date=$(date +"%Y-%m-%d_%H-%M-%S")
cp *.txt backup/
echo "Backup completed"
```

**OUTPUT:**

```
/drives/c/Users/Qamar Raza/OneDrive/Desk
2026-03-15  00:12.19  ./assign7.sh
Backup completed
```

To confirm

```
/drives/c/Users/Qamar Raza/OneDrive/Desktop/NucleusTeq Training/2.Linux
2026-03-15  00:18.48  ls
Linux\ Assignments.docx        assign7.sh  bio.txt        my_file.txt     script.sh  task2.sh  task4.sh
Linux\ Training\ Document.docx backup      linux_practice port_check.txt  task1.sh   task3.sh  ~$nux\ Assignments.docx
```

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----------END OF ASSIGNMENT----------